

Career Recommendation for students

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Article Info

Article History:

Published: 22 April 2026

Publication Issue:

Volume 3, Issue 4
April-2026

Page Number:

2054-2063

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Abstract:

In today's dynamic job market, college students usually find it difficult to identify career paths that align with their respective skills and aspirations. This creates a mismatch in the decision of career choices due to a lack of informed decisions. A Machine Learning-based Career Recommendation System using the K-Means clustering algorithm is designed, along with Natural Language Processing-NLP. This system analyzes students in respect of their educational backgrounds, programming skills, usage of tools, and interests in various careers, aside from NLP-based extraction of skills from resumes. Based on these, it recommends the most compatible career options that fit the student's skill set and also align with their long-term career goals.

Keywords: ML, Career Interest, Recommendation system, K-means, NPL, Roadmap, Resume Extraction, Chatbot

1. Introduction

Choosing a career is difficult for this generation of students. Even when their interests and abilities are recognized, the majority of students lack the knowledge necessary to make wise career decisions. Some of them ask seasoned professionals for assistance on jobs. These suggestions, meanwhile, might only be based on the experts' current skill profiles, past experiences, and industry trends.

Students are consequently unable to overcome their doubts and choose a career path that effectively fits their abilities and long-term goals.

This is provided in order to solve the previously noted issue with the Career Recommendation System for Students. When suggesting a career path, the student's academic background, program skills, familiarity with technology and tools, and professional goals will all be considered.

Using a real dataset, the system collects information about students' actual employment preferences.

One machine learning method that is used in conjunction with natural language processing (NLP) to extract text from student resumes and determine their employment interest category is the K-Means clustering algorithm.

The algorithm predicts suitable career roles and recommends two or three other jobs based on the students' skills, allowing them to weigh their options and decide on the best course of action.

The approach also provides a road map for helping students acquire the required skills after they have been selected.

This strategy is used because it gives pupils a suitable route to long-term goals and lessens career uncertainty.

2.Literature Review

Numerous researchers have worked on career suggestion systems to help students make the right decision. Various machine learning and data mining techniques have been applied to forecast job interest based on knowledge and educational background.

The authors [1] discussed a student's career interest and created a chatbot that uses machine learning and natural language processing (NLP) technology to offer job advice. The author employs a chatbot and the machine learning K-Means algorithm to direct students' career interests. But when it comes to dynamic and customised career counselling, the chatbot is complicated.

There are a lot of online career recommendation tools, according to the researchers [2]. Because this recommendation system was not created with computer science graduates in mind, they may obtain suggestions for professions unrelated to their degree. The student career recommendation system makes use of a content-based screening method.

The [3] recommendation system, which classifies IT skills to determine the optimal career path, was developed by students studying informatics engineering. Students will also be able to get information from IT experts through their LinkedIn accounts and utilise it for reference.

The System [4] is incredibly basic, extracting the required abilities through an open pipeline. However, the approach just shows the talent gap and does not offer a precise career recommendation.

This System [5] assists high school students in selecting a field of study and a potential career by looking at their character and academic qualities. Since engineering is one of the most popular subjects among high school students, this paper constructs higher education in the field. According to the computer exam's findings about students interests and academic preferences.

A system that proposed a stream from the arts, business, or science is provided by the authors [6]. There were two parts to the academic knowledge and aptitude test. The Random Forest regression method is used to provide students with accurate test performance suggestions. Students should make a professional decision as soon as possible.

According to the report's author [7] at an early age, students are expected to make informed judgements about their future. There is an urgent need for an automated, intelligent method that can successfully mentor kids.

The study of the paper [9] investigates the use of ML methods in predicting student of academic achievement and career placement aspect the authors given the spitted of the growing adoption of predictive analytics within education in the majority of university available systems lack certain evaluation.

The paper [10] provides the technology driven solution need of student and recommend in accordance with their interests, academic streams and aptitudes. This project paper proposed a career recommendation system for needs 10th grad students. The system take input from user detailing subjects of interest, stream (science, commerce & Arts), hobbies, and personality. The system uses this input to recommended potential future career options.

This paper [11] authors used decision trees to help classify student into who Drop the education in middle way and choose another career path using machine learning model.

The authors [12] recommender's system in a computer program builds with the help of expert where the details of the students and their aptitude help finding a right course for the future. Choosing a right field in (CSE, IT) stream is very important for their future.

The paper [13] examines authors provided to student career-based project recommendation system aims to provide personalized project recommendations based on the user's responds to a series of questions about skills, interests and career aspiration

The paper studies [14] a career recommendation method for college student based on occupational value. The study provided a detailed explanation of the concept of generating career suggestion results for college students based on occupational value. It also used experimental data to verify the correctness of the recommended profession guidance.

The study [15] looks at the Learning Environment Recommendation System, which makes recommendations for students' courses, majors, specialisations, and even jobs. The recommendation system may also have many uses. By assisting students in selecting a course or major, these systems can recommend a career path based on their learning style, interests, and skills. Considering important factors that affect a student's future career.

The study's authors [16] may recommend a career choice for the student. Machine learning algorithms such as KNN, Support Vector Machine (SVM), and Random Forest predict career interest based on input including extracurricular activities, academic strength, personal interest, and part-time work experience.

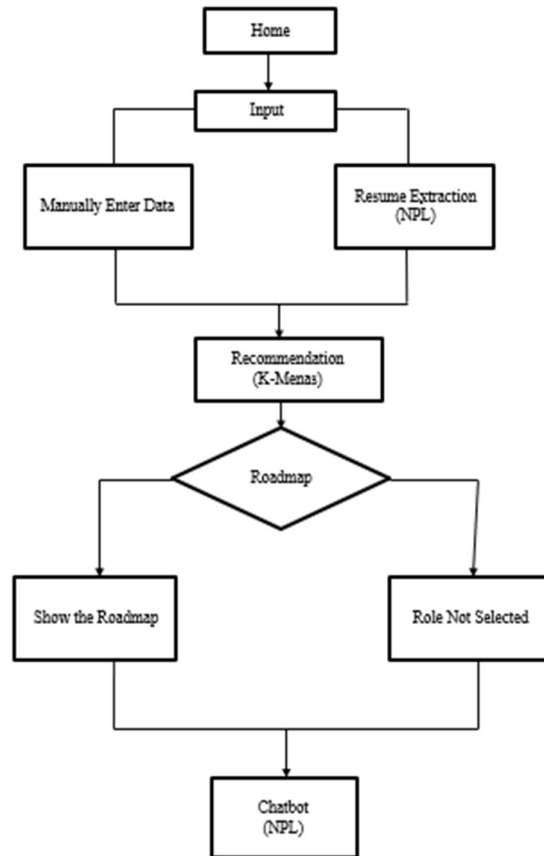
3.Existing system

Minimal emphasis on academic performance in order to provide job alternatives, the majority of current career advising systems mostly rely on students' academic scores. The recommendations frequently don't match the student's potential or desire, and they don't take into account other crucial elements including abilities, hobbies, soft skills, personality type, and long-term objectives. Confusion and poor career decisions result from this. Absence of a Customised Roadmap The majority of existing systems only provide a list of careers, but not how to go about pursuing them It does not plan out in detail for the student a step-by-step approach regarding required courses, skill training, internships, certifications, future job roles, salary expectations, and growth opportunities. Definitely, without this roadmap, it is hard for the students to take further action.

4.Proposed System

The Proposed Career Recommendation System is designed to guide Students in Choosing the right career path based on student input like academic detail, skills and interests. The architecture considers a following major component.

Fig 1: System Architecture



4.1: System Architecture

The above Fig 1 Shows that project Architecture

1. Home: The system begins with home interface where user can start the career recommendation process and navigate to different pages of the application.

2. Input: The Systems collect the student details in two methods:

Manually Enter Data: Student Provide their career information like education background, skills and interests career role.

Resume Extraction: The system automatically reads the resume and extract important information like academic details, related skill using text processing techniques.

3. Recommendation: The suggestion After the data is gathered, a machine learning model is used to process it. After analysing the dataset, the recommendation engine matches student-related fields based on the best career choice.

The engine suggested two or three different job possibilities that fit the student's skill set.

4. Roadmap: Diagram The student made their choice based on the roadmap that included both text and an image.

Two possible results:

- **Role selected:** A chosen role The chosen role's roadmap is being provided by the student.
- **Role Not selected:** The student is content with their assigned function and is developing new abilities.

5. Chatbot: In order to speak with the chatbot, the student clarifies their query. The student poses a query about their career, and the chatbot responds in response.

4.2: Implementation

The proposed system Implements as web application using Steps

A. Data Collection: To collect the student real dataset involving student academic details, skills and career interest using Google form and covert CSV file

- **CSV File:** Common Separate Value (CSV) file containing Rows of the student record

B. Data Preprocessing: Load the Google form into Excel to clean the dataset to handle the missing values, redundancy data and standardized data, auto-reconstruction.

C. Web Design: The website design using the frontend and backend technologies

- **Frontend:** The frontend of the system is developed by using:
HTML – Defines structure and user interface screens
CSS – Provides responsive and styling for website
JavaScript – handling Interaction UI behaviour
- **Backend:** The backend is implements using Python with Flask framework to perform the handling user requests and uploading files, uses Routing navigate to one page to another

D. Database: The SQLite Database stored the student details like skills, education background and interest career what are student give input fields

E. Machine Larning Algorithms: The machine learning is subset of AI train the model using data and method without exploiting the programming.

I. K-Mean: K-Mean is papular unsupervised machine learning algorithm used to group similar data point into cluster. The K-Menas group the student who have similar skill and career interestscustering into group.

II. Natural Language Processing (NLP): The NLP is filed of AI that helps computer understand, interpret and generate human language.

The NLP read the resume and Extract text into resume to required filed.

4.3: Result and Discussions

The developed career recommendation for student implemented using k-Menas for clustering and Natural Language processing technique for text extract from resume. The system was tested with real student dataset to evaluate to performance in predicating suitable career roles. This project developed several components:

Home Modulo

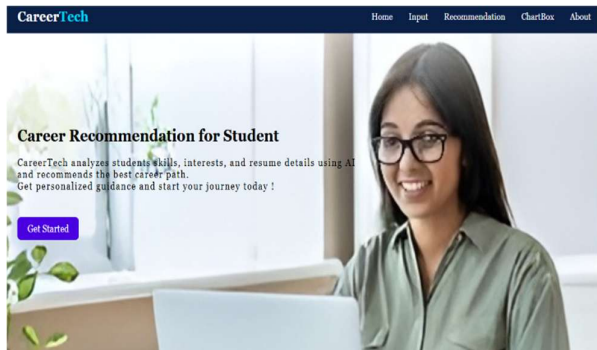


Fig 2: Home page

The home page is staring entry point of the project. It provides the user-friendly interface where student can start the project. This project including navigation such that Home, Input, Recommendation, Chatbot and about are placed at top of the menu button and have Get stated button to start the project.

Input Modulo:

Fig 3: Input page

Input Page the system collects the student data in two possible methods:

Manually Enter Data: Student Provide their career information like education background, skills and interests career role.

Resume Extraction: The system automatically reads the resume and extract important information like academic details, related skill using text processing techniques.

Recommendation Modulo:



Fig 4: Recommendation page

Recommendation page Once the input is collected, its processed using machine learning model After analysing the dataset, the recommendation algorithm matches student-related fields to determine the best career choice. Based on the student's skill set, the algorithm recommended two or three potential job options.

Roadmap Modulo: Based on the roadmap, which featured both words and an image, the student made their decision.

Two possible outcomes:

Role selected: The chosen role's roadmap is provided by the student. The student provides the roadmap for the selected responsibilities.



Fig 5.1: Role selected

Role Not selected: The student is learning new skills and is happy with the role they have been given.

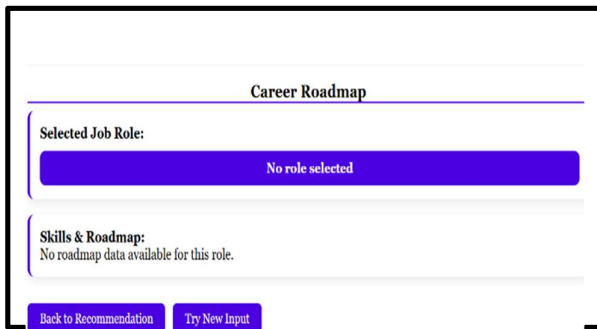


Fig 5.2: Role not selected

Chatbot Modulo:

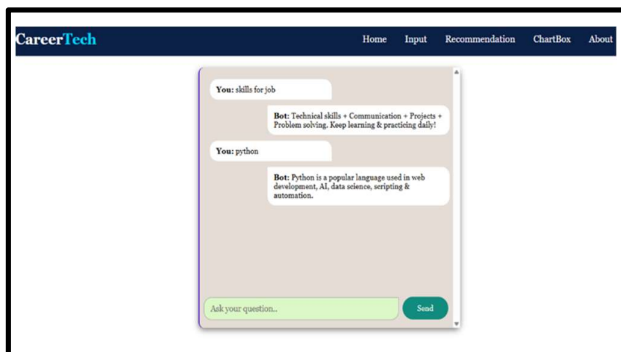


Fig 6: Chatbot page

The student clarifies their question to communicate with chatbot. The student asks their career related question and chatbot answer based on asking the question

About Page:

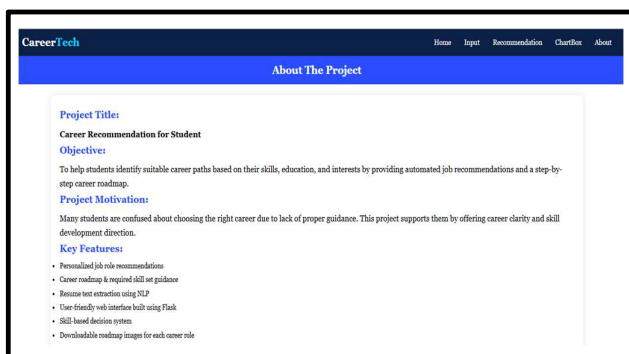


Fig 7: About page

The about page consider information of the project user can understand the actually they project doing and how help the student get their career related role.

5. Conclusion

The purpose of the career recommendation is to assist students in choosing the right career. decision based on the talents, career preferences, and academic information of the student. Natural Language Processing (NLP) and the machine learning method K-Means are used to efficiently read the text and K-Means clustering students according to similar talents and career goals, as well as extracting text from resumes.

Along with a thorough career path that leads the student step-by-step to the chosen role, it creates individualised employment role recommendations.

This project makes bridges the gap between student's abilities and the growing demands in the IT industry by providing actionable thought, required skills set, and learning path.

Overall, this project contributes to a more structured, informed and skill-based career planning approach, making a valuable tool for academic and professional growths.

Future Enhances

The system can link the job portal directly; this help student understands which career have high demand and growth opportunities.

In future can include the personalized learning path recommended courses, certificates and attending the workshop to understand the current job market.

To build a Mobile app that student can help access the system anywhere and anytime.

Make the student profile and get notification in mobile for take the test for their knowing skills it helps student to revise the particular topic.

The system can recommend internships, academic projects, and extracurricular activities based on career goals.

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